

Maturing Mars Small Satellite Mission Concepts to Inform Bounds for Delivery Tug

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Abstract

The mainstream exploration of Mars has been previously limited to flagship missions, containing instrumentation to fulfill many science objectives. Martian exploration has now turned to more dedicated and low-cost missions. To maintain small satellite size constraints, a delivery tug will provide propulsive power to arrive at Mars, and serve as telecom relay support to Earth. Two small satellite concepts to inform the design bounds for the tug and enter a menu of missions to send to Mars are: a gravity measurement orbiter projected to be half the mass of its analogous mission, GRAIL, and an in-situ weather monitoring hard lander that will produce continuous measurements with possible networking capabilities with other landers or orbiters. Both concepts have been robust to changes in delta v and thermal needs providing flexibility to the mission overall, and will be able to perform decadal level science in line with the Mars exploration plan for the next 20 years. These two mission concepts along with others in a strawman mission, will further narrow the drop-off orbit, telecom needs, and mass capacity of the space tug to bring the concept closer to a point design and prove the viability of dedicated small missions on another planet.

Introduction

With the steadily increasing cost of the Mars Sample Return (MSR) Project, the Mars Exploration Program office has proposed a method of spending that promotes a more predictable and even cadence that can be upheld year after year rather than the spikes and drops exhibited by large flagship projects¹. In order to do that, as well as promote international Mars exploration collaboration, JPL is a participant in the International Mars Exploration Working Group (IMEWG) as well as the Low-Cost Mission Working Group (LCMWG) within IMEWG. This Low-Cost working group is conducting the Earth2Mars study where member nations are generating low-cost mission concepts within ESPA-Class SmallSat (<400kg), CubeSat (up to 12U), or hosted instrument mass classes that can be transported to Mars using a hypothetical space tug. The constraints for the concept include a 300 km altitude circular polar drop off orbit, mass class constraints described earlier, and the ability for standardized attachment to the space tug which could look different based on the mass class. In addition, it would be assumed that the tug provides

propulsive needs for Mars orbit insertion and will provide support as a Direct to Earth (DTE) relay for these missions. These low-cost, smaller missions have the objective of performing decadal level science at their size in focused mission profiles. In order for that to happen, these concepts will be considered more highly if they have high science precedence and high feasibility which is delineated for a number of concepts in the MCE-SAG Final Report in a grid of those two categories². In addition, it requires little technological advancement if there is sufficient heritage for the instrumentation in the correct mass range that allows for a small vehicle to house it.

Based on these parameters, a gravity science mission and a weather monitoring mission provide substantial data to the science community across many science facets including atmospheric science and climate science as well as providing building blocks for eventual human exploration. Even further, gravity science at Mars would contribute heavily to geological sciences and give insights to Mars' past^{1, 3}. These concepts and more also contribute to the framing of accommodations that the space tug must offer in order to ensure mission success at the lowest cost. Accommodations such as drop off orbit and final orbit are most impactful to the concepts directly, but more indirectly, it produces options in propulsion techniques and relay considerations in terms of view of orbiters and whether or not there are hosted instruments. Ultimately, it results in a viable path for Mars science to move forward in a manner more similar to Earth's slew of small satellite missions providing a low barrier of entry for an international effort to study Mars as a system.

Results: Overview of concepts

Concept 1: Gravity Measurement Orbiters

Science

The gravity measurement orbiters concept is one that showed high feasibility because of the deep heritage given with satellites such as GRACE, GRACE-FO, and GRAIL which orbited the Earth, Earth again, and Moon respectively, and high precedence because of their impact on an in depth understanding of the mass make-up of Mars^{2, 3, 4}. This mission would fulfill the science goal of determining the global structure of the martian crust and lithosphere, including the origin and nature of the hemispheric dichotomy². The missions conducted at Earth achieved more than simply a mass distribution of the planet based on mass and gravity, it was able to reveal water behavior throughout the planet⁵. It is given by the Keck institute of Space Sciences that gravity data is one of the most high-precedence measurements for understanding the revolution of rocky planets⁶.

Subsystem breakdown

Modeled in the JPL TATER tool for systems engineers, a SmallSat gravity orbiter utilizing a modified version of the Lunar Gravity Ranging System (used on GRAIL mission) is capable of fitting within the smallsat format when considering the usage of smallsat sized attitude control

systems, telecommunications, and command and data handling systems. The orbit insertion performed by the delivery tug contributes to massive mass savings in propellant. Both the fuel mass and the mass of the structure itself. In addition, the model exhibits a 71% drop in predicted wet mass from GRAIL's flown values further emphasizing the benefits given by a delivery tug. Explorations for use of the CubeSat-sized flight computer Sphinx (also used on Lunar Flashlight⁷) as well as the IRIS Radio should be conducted after the results of this paper.

Connops

During the mission lifetime, the orbiters will experience three key steps in the nominal design. The drop off and maneuver to science, the science mission itself plus maneuver periods throughout, and the end of life. The drop off orbit considered for the study is very close to that of the desired science orbit, maybe differing by a few degrees. This potential plane change would be accompanied by an altitude drop in order to travel to a very low Mars polar orbit with an altitude of 200 km in order to obtain the final science orbit. Science operations will be continuous for a 30-50 Earth-day cycle before shutting off to perform correctional maneuvers and desaturation, mimicking the operational procedures of GRACE and GRAIL⁸.

The power modes considered for this vehicle were checkout, science, maneuver, telecom, and battery post-deployment in order to capture the driving power mode of checkout where all power systems were on. For end of life considerations, the orbiters could transfer to a benign orbit or crash into the surface of Mars in an unprotected location.

Concept 2: Dust and Weather Hard Lander

Science

The dust and weather hard lander concept would contribute to the ongoing data collection of global weather on the surface of Mars. With the rovers on Mars hosting their own instrumentation suites for meteorological analysis, a focused mission would be able to continuously collect and transmit that data without other science needs having to be met. In addition, the dust activity on Mars has led to much literature and analysis prompting further focused study in order to understand atmosphere/climate variability including the timing of dust storms².

Subsystem Breakdown

JPL has previously matured the concept for a hard impact lander called SHIELD⁹ within which there is an accordion-like structure that houses instruments up to 6 kg. The SHIELD vehicle also consists of a Warm Electronics Box (WEB) capable of hosting instruments up to 120 W between -20 and 0 °C accounting for all technology needs aside from the instruments that must survive the 70 m/s impact into the Martian surface and survive the bounce that may occur after. That is also inclusive of the solar panels used to power the WEB that are on the outside of the

vehicle. The instruments also have a soft size limit of 30 cm x 30 cm x 10 cm in their stowed configuration⁹ which is treated as a hard limit for this concept.

Connops

The science mode for this instrument is continuous once calibration has been achieved. Because there is no imager on board in the current design, the data sent to the relay tug will be manageable, even while assuming the tug is dealing with data from all other missions on the manifest in a limited timeframe. There is a battery within the WEB in case of dust events, and the lander is designed to survive the Martian night and winter for up to one Mars year within 30° North and 30° South in latitude⁹. This vehicle was modeled in TATER in order to better understand the subsystems of the vehicle, and the modeling resulted in reported model mass values found previously for SHIELD, increasing the confidence in this “visualization” model.

Methods: Trade Studies to Create Detail and Trades Analyzed

The modeling software TATER is used by system engineers at JPL in order to perform initial studies of a concept and obtain high level mass rollups from past missions and mission concepts to drive expectations and mature mission ideas^{10, 11}. This software was used to model the nominal values for each concept. TATER is only verified for element level mass values (total dry predicted and total wet allowable) but can be helpful in “sanity checking” the values when trading values and visualizing subsystems and options because of the desire to model small satellites. This provides the primary methodology for maturing the concepts up to this point.

Concept 1: Gravity Orbiter

Thermal Power

Thermal power was considered as a worthwhile trade for the gravity orbiter in order to determine the bounds of thermal power capabilities while cold-biasing the spacecraft, as well as visualizing the sensitivities of absorptivity and emissivity combinations when attempting to obtain a specified steady state bus temperature. This trade was considered because of the sensitivity to temperature that gravity orbiting instruments have because of the spacecraft to spacecraft pointing accuracies necessary to return usable science¹². Two possible internal power needs were modeled to see how they would affect mass. These were chosen by modeling the absorptivity and emissivity along x and y axes, and generating a contour that represented the thermal heating power necessary to reach a steady state temperature of 10 °C in a single-node analysis. Two graphs were generated, one for the maximum power bound represented by the farthest distance from the sun with the lowest internal power mode, and one for minimum power represented by the closest distance to the sun and highest internal power to ensure that cold biasing was occurring. The conclusions made from this were that the absorptivity and emissivity pairings were quite sensitive along the contour jumping from 80 or so Watts to 140 or so Watts of necessary thermal power when the absorptivity

decreased by 0.01 and the emissivity increased by 0.03. In addition, the mass distributions changed significantly with a 28 W decrease in thermal power prompting an 8% decrease in overall dry predicted and overall wet allowable mass.

Delta v capacity

Delta v capacity is a major consideration for space vehicles and even more so for ones traveling outside of Earth's orbit. This propulsive limiter on smallsat mass is one of the major reasons the space tug concept is so important for enabling small satellite research at Mars¹³. Despite this propulsive assistance, plane changes in particular are still very expensive in delta v. Modeling the mass changes that occur when delta v capacity is increased is crucial to remaining within the allotted smallsat mass budget of 400 kg. A logarithmically scaled increase in mass (y axis) from the nominal mass was modeled against the delta v needs (x axis) including an altitude drop from the expected drop off orbit of 300 km to the desired science orbit of 200 km in order to determine reasonable delta v capacities that can be obtained. This analysis showed that the science inclination has limited range when compared to the inclination of the drop off orbit. Even a 5 degree plane change ends up outside of the mass limit which was about 1150 m/s. This mass budget would also include considerations an engineer must make for stationkeeping and potential correctional maneuvers, so whatever ends up being the delta v budget, even less can be used for orbit changes. This may drive science orbit limitations for the tug and enforce drop off orbit needs if there is a large range of desired orbits for missions on a specific tug manifest.

Concept 2: Dust and Weather Hard Lander

Instrument suites

The topic of instrument suites is considered a trade in this concept because of the versatility of meteorology, dust, and weather instruments. They tend to be fairly small with low power needs and low mass and are able to be "mixed and matched" because they operate independently from one another. Given this, different devices were traded with one another in order to determine that a temperature sensor, pressure sensor, and wind speed instrument in order to fulfill the science objectives, and other sensors would be very easily integrated into the system, but serve as nice-to-haves.

Configurations

Since the SHIELD vehicle provides a 30 x 30 x 10 cm space to house stored instruments, deployables, and potential extra equipment, The configuration of those objects, as well as placements of deployable booms or arms was important in determining the feasibility of this concept. Various instrument suites were mapped into this spatial area in order to prove that they could fit, plus provide extra room for electronics boxes, padding, and deployables. From this analysis, it was found that a slew of meteorological instruments could fit inside of this

configuration, but the tallness of some got in the way^{14, 15}. To make up for the size, things like wind sensors and relative humidity sensors could be put on extendable booms that must survive the 70 m/s impact into the Martian surface. In order for the instruments to operate continuously, there is also room for extra batteries if needed to survive the winter or night. The main conclusion of this study was that a focused meteorological lander is very feasible in this space and even leaves room for extra equipment if the need arises throughout further study and concept maturity.

Discussion: Concluding thoughts for delivery tug and areas for future work

Both of these missions are highly enabled by the propulsive cost being taken up by the delivery tug in addition to burdens of data relay to Earth. These concepts also help to uncover the accommodations that must be provided by the tug in order to make the mission successful. Propulsion considerations for the tug itself offers the choice between chemical, Solar Electric Propulsions (SEP), or a hybrid. Chemical provides a shorter cruise duration and higher thrust, but will most likely be a higher mass and low efficiency. SEP would be higher efficiency and lower mass, but lower thrust and a longer cruise duration, contributing to the decay of the satellites being carried before they have even started their missions¹⁶. Hybrid propulsion systems have a mix of the pros and cons that must be evaluated for further consideration. These propulsive types may also contribute to the considerations for data relay. If SEP is utilized, the large solar panels necessary for traveling to Mars could serve a double purpose and provide substantial power for the tug during the relay orbit phase of the tug's life. Even the consideration for a final orbit for the tug is dependent on the missions on the tug itself because of the option for hosted instruments on the tug and their needs for science. The tug could potentially remain in the final drop off orbit, expand to a higher orbit for more coverage, or move to a different science orbit for the hosted instruments.

In addition, the drop off sequence is an area that is affected by the tug manifest. There are many high level options inclusive of the tug performing similarly to a public bus where it makes stops at or near each orbit needed for science and deploys, the usage of another cubesat deployer on board that provides the orbit changes necessary for cubesats and potential hosted instruments, or a singular stop decided upon based on the mission manifest that will require all missions to provide their own delta v in order to reach their desired science orbits. Given all of these options and the mission potential with a carrier vehicle like this, it would really depend on each manifest for which option is ideal. Tug mission manifests may be restricted to one particular orbit or mission type, or to a particular theme depending on the flexibility and capabilities of the tug.

Conclusions

This type of mission profile with focused concepts could enable multiple decadal level science objectives being reached by one large launch being dispersed to different orbits. The presence of the tug lowers the barrier of entry for science to be done at Mars enabling higher returns and more discoveries to be made with successful execution. This would then showcase the beginning of a mimicry of earth science style missions at Mars. With earth science conducting numerous successful smallsat missions at a steady cost point that is lower than flagship missions, it provides a good model for what can be done at Mars, keeping in line with the most current 20 year plan. The space tug would enable that with the goal of becoming a standardized vehicle built with the recurrence of missions in mind; so another can be built with a different distribution of missions as manifests develop and science progresses.

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